

## Physics layer $\Phi$ (frozen)

self-supervised dynamics · behavior proposals

*trained once, reused across taxonomies*

embeddings  
+ proposals

## Semantic layer $\Omega$ (revisable)

anchor-guided pseudo-labeling · self-training

*taxonomy evolves  $\mathcal{Y} \rightarrow \mathcal{Y}'$*

**only  $\Omega$  retrains**

budget retention · own full-budget top-1 kept at 10% labels (canine: 9.8 vs 11.1% chance at 13% budget)

100%

NTU60

90.7%

NTU120

88.9%

canine

28.9%

